

Q. I have a Raspberry Pi model B board and plan to access the Internet through Wi-Fi adaptor. I am using the latest Raspbian image. Please help me set up the wireless adaptor.

Kiran Sabne
Through e-mail

A. In most cases you would need an Ethernet connection in the first place to set up your Wi-Fi adaptor.

Connect the Ethernet to the Raspberry Pi board. Normally, you should immediately get access to the Internet. The only setting that you would need is your router which should be configured for DHCP. You can check the Internet connection by browsing some website with any of the browsers already available with the Raspbian. If the Internet is through, follow these instructions:

1. Connect your wireless adaptor using a USB powered hub. This will save your Raspberry Pi from wear and tear.

2. Open LxTerminal and type the below command to find out the name of the device and the relative drivers required:

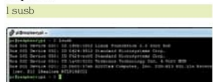


Fig. 1: lsusb command

In the particular case shown in Fig. 1, the fifth item detected is the wireless adaptor. And the drivers required would be Realtek RTL8192CU.

Now you have to look for these drivers and install them if not installed already.

Please note that your drivers will be different.

3. Before proceeding further, update your package information using the following command:

`sudo apt-get update`

4. Now install the required driver by:

`sudo apt-get install firmware-realtek`



Fig. 2: Install drivers

Replace firmware-realtek with the driver required for your wireless adaptor.

5. Once the driver is successfully installed, use the following command to confirm:



Fig. 3: lsmod

The ninth item shows the driver 8192cu installed.

6. Once the drivers are properly installed, open Wi-Fi configuration utility, which comes preinstalled with Raspbian.



Fig. 4: Wi-Fi configuration

This utility is not available in older versions of Raspbian. So get the latest one installed in your SD card.

7. Click 'Scan.' Another window will open. Find your network access point in the list. Double-click your wireless access point.

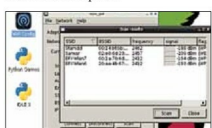


Fig. 5: Wireless access point

8. Another window will open. Put your password in the PSK field and click 'Add.'

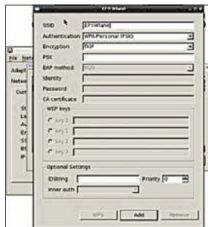


Fig. 6: Enter password

9. Close everything and get back to the initial Wi-Fi configuration window. Click 'Connect' and you should see it connected. This window also shows the IP of the Raspberry Pi allotted by the router. You will need this IP for remote access.



Fig. 7: Connection established

10. The ifconfig command will also indicate the same connection.

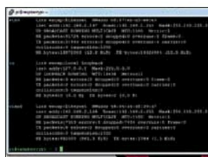


Fig. 8: ifconfig

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efyindia.com) and should include name and address of the sender